

Enterprise VPN, Firewall & Secure Access Implementation

NETWORK ADMINISTRATION INTERNSHIP PROGRAM



NETWORKING

ITSimplera Institute

Week 07 Project Report

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Chapter 1 Introduction

1.1 Background

This report documents Week 7 of the Network Administration Internship Program at IT-Simplera Institute, undertaken under the supervision of Jawad Qayum, Senior Network Administrator. Building on the segmented, service-rich enterprise network established in earlier weeks, Week 7 shifts focus from internal service delivery to perimeter and remote-access security. The assignment simulates a two-site enterprise — a Headquarters (HQ) office and a Branch office — connected over an untrusted WAN link, and requires that all traffic between them be encrypted, inspected, authenticated, and monitored.

The project was implemented in GNS3 and covers five security domains: Site-to-Site VPN using IPsec, perimeter defense using a Zone-Based Firewall (ZBF), secure administrative access using AAA and SSH, routing-plane security using OSPF authentication, and centralized monitoring using Syslog, NTP, and SNMPv3. Together these controls model the layered defense-in-depth approach expected of a production enterprise network.

1.2 Objectives of the Assignment

- Implement secure branch-to-branch communication using IPsec Site-to-Site VPN technology.
- Configure and verify Zone-Based Firewall (ZBF) policies to protect enterprise resources.
- Deploy AAA authentication and secure remote management using SSH.
- Implement device hardening and secure routing practices (OSPF authentication, passive interfaces, prefix filtering).
- Monitor, verify, and troubleshoot security configurations using Syslog, NTP, and SNMPv3.

1.3 Scope of the Report

This report is organized into eight chapters. Chapter 1 introduces the assignment and defines the key security terminology used throughout. Chapter 2 profiles IT-Simplera Institute and the Network Administration department. Chapter 3 details the two-site network topology and addressing scheme. Chapter 4 documents the configuration of each security control — VPN, firewall, secure management, routing security, and monitoring. Chapter 5 presents verification and connectivity testing, Chapter 6 covers troubleshooting, Chapter 7 summarizes tools and skills applied, and Chapter 8 concludes with lessons learned.

1.4 Key Terminology & Definitions

The following terms and protocols form the technical foundation of this assignment and are referenced throughout Chapter 4.

VPN Security

Term	Definition
IPsec	A suite of protocols that authenticates and encrypts IP packets exchanged between two hosts or gateways, providing confidentiality, integrity, and origin authentication over an untrusted network.
Site-to-Site VPN	A permanent, gateway-to-gateway IPsec tunnel that transparently encrypts all traffic between two office LANs without requiring client software on end hosts.
ISAKMP / IKE Phase 1	Internet Security Association and Key Management Protocol — negotiates a secure, authenticated channel (the IKE Security Association) between two peers before the actual data tunnel is built.
IPsec Phase 2	Negotiates the IPsec Security Associations (SAs) that protect the actual data traffic, using parameters agreed in Phase 1.

Term	Definition
Pre-Shared Key (PSK)	A shared secret string configured identically on both VPN peers, used to authenticate each other during IKE Phase 1.
Transform Set	Defines the combination of encryption (e.g. AES) and hashing/authentication (e.g. SHA) algorithms applied to protected traffic in Phase 2.
Crypto Map	A router configuration construct that binds a peer address, transform set, and interesting traffic (via ACL) together and applies the resulting policy to an outbound interface.

Firewall Security

Term	Definition
Zone-Based Firewall (ZBF)	A Cisco IOS firewall architecture in which interfaces are assigned to security zones, and traffic is inspected or blocked based on policy applied between zone pairs rather than per-interface ACLs.
Security Zone	A logical grouping of one or more router interfaces that share the same trust level (e.g. ZONE-INSIDE, ZONE-OUTSIDE).
Zone Pair	A one-directional relationship between a source zone and a destination zone to which a firewall policy is applied.
Class Map	Defines the traffic (by protocol, ACL, or other match criteria) that a policy will act upon.
Policy Map	Associates a class map with an action — inspect, pass, or drop — and is applied to a zone pair via a service policy.
Service Policy	The command that attaches a configured policy map to a zone pair, activating the firewall inspection rules.

Secure Device Management

Term	Definition
SSH Version 2	An encrypted remote-management protocol that replaces plaintext Telnet, protecting credentials and session data from interception.
AAA	Authentication, Authorization, and Accounting — a framework for controlling and logging administrative access to network devices, backed here by a local user database.
Login Banner (MOTD)	A message displayed before login that provides a legal warning against unauthorized access.
Console / VTY Access Control	Restricting and password-protecting the physical console line and virtual terminal (remote) lines used for device management.

Routing & Monitoring Security

Term	Definition
OSPF Authentication	MD5 or SHA authentication of OSPF hello and update packets, preventing rogue routers from injecting false routes into the routing domain.
Passive Interface	An interface on which OSPF listens but does not send hello packets, preventing adjacency formation on LAN segments with no other routers.
Prefix List	A route-filtering mechanism that permits or denies specific network prefixes from being advertised or accepted, more efficient than an ACL for routing filters.

Term	Definition
Syslog	Centralized forwarding of device event, warning, and error messages (UDP 514) to a logging host for audit and troubleshooting.
NTP	Network Time Protocol — synchronizes device clocks to a common reference so that Syslog and SNMP events can be correlated accurately across the network.
SNMPv3	The latest SNMP version, adding user-based authentication and encryption (in contrast to plaintext community strings in v1/v2c), used here to securely poll device health.

Chapter 2 Organization Profile

2.1 About the Organization

IT-Simplera Institute is an IT training and solutions organization that runs a structured Network Administration Internship Program, providing interns with practical, lab-based exposure to enterprise networking and security concepts through weekly, progressively advanced assignments delivered in Cisco Packet Tracer and GNS3.

2.2 Department / Team

The internship is carried out within the Network Administration department, under the direct supervision of Jawad Qayum, Senior Network Administrator. The department sets weekly technical assignments, reviews submitted documentation and lab files, and conducts evaluation interviews on Google Meet to assess each intern's understanding, implementation quality, and problem-solving ability.

2.3 Assignment Context

Week 7 moves the internship from single-site service deployment (Weeks 4–5) to a two-site secure enterprise model. A Headquarters router and a Branch router, each fronting a local LAN, are connected across a simulated WAN and secured end-to-end using the controls listed in Section 1.4.

Chapter 3 Network Topology & Security Design

3.1 Project Overview

The lab models two sites — Headquarters and Branch — built in GNS3 as two IOS routers connected back-to-back over a point-to-point WAN link. Each router fronts a local-area network reached through an access-layer switch (R3 at Headquarters, R4 at Branch) and a single VPCS test host (PC1 at Headquarters, PC2 at Branch). An IPsec Site-to-Site VPN tunnel is built across the WAN link, a Zone-Based Firewall is applied on the Headquarters router to inspect traffic between its internal zone and the WAN-facing zone, and OSPF with MD5 authentication is configured on the Headquarters side of the link.

3.2 Network Topology Diagram

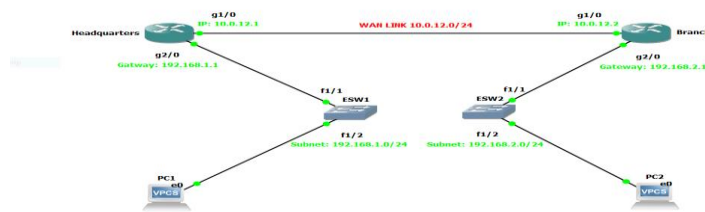


Figure 1: GNS3 topology — Headquarters and Branch routers, R3/R4 access switches, PC1/PC2 VPCS hosts

3.3 IP Addressing Scheme

Device / Segment	Interface	IP Address	Role
Headquarters WAN	Gi1/0	10.0.12.1 /30	VPN peer / ZBF outside / OSPF
Headquarters LAN	Gi2/0	192.168.1.1 /24	HQ gateway / ZBF inside
Branch WAN	Gi1/0	10.0.12.2 /30	VPN peer
Branch LAN	Gi2/0	192.168.2.1 /24	Branch gateway
PC1 (VPCS)	e0	192.168.1.0/24 range	HQ test host, via switch R3
PC2 (VPCS)	e0	192.168.2.0/24 range	Branch test host, via switch R4

3.4 Security Zones

Zone	Member Interface	Router	Trust Level
INSIDE-ZONE	Gi2/0 (192.168.1.0/24)	Headquarters	Trusted internal LAN
OUTSIDE-ZONE	Gi1/0 (10.0.12.0/30)	Headquarters	Untrusted WAN-facing

Chapter 4 Security Implementation & Configuration

The configuration below is transcribed directly from the latest show running-config on both routers. Screenshot placeholders mark where CLI verification output should be inserted.

4.1 IPsec Site-to-Site VPN

IKE Phase 1 (ISAKMP) policy, a pre-shared key, a Phase 2 transform set, and a crypto map are configured on both Headquarters and Branch to build an encrypted tunnel across the 10.0.12.0/30 WAN link.

```
! Headquarters
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 5
crypto isakmp key cisco123 address 10.0.12.2
!
crypto ipsec transform-set MY-TRANSSET esp-aes 256 esp-sha-hmac
  mode tunnel
!
crypto map MY-CRYPTO-MAP 10 ipsec-isakmp
  set peer 10.0.12.2
  set transform-set MY-TRANSSET
  match address VPN-TRAFFIC
!
ip access-list extended VPN-TRAFFIC
  permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
!
interface GigabitEthernet1/0
  ip address 10.0.12.1 255.255.255.252
  crypto map MY-CRYPTO-MAP
```

```

! Branch
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 5
crypto isakmp key cisco123 address 10.0.12.1
!
crypto ipsec transform-set MY-TRANSSET esp-aes 256 esp-sha-hmac
  mode tunnel
!
crypto map MY-CRYPTO-MAP 10 ipsec-isakmp
  set peer 10.0.12.1
  set transform-set MY-TRANSSET
  match address VN-TRAFFIC
!
ip access-list extended VN-TRAFFIC
  permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
!
interface GigabitEthernet1/0
  ip address 10.0.12.2 255.255.255.252
  crypto map MY-CRYPTO-MAP

```

```

Headquarters#
Headquarters#
Headquarters#
Headquarters#
Headquarters#
Headquarters#
Headquarters#show crypto isakmp sa
dst          src          state          conn-id status
10.0.12.2    10.0.12.1    QP_IDLE        1801 ACTIVE
IPv6 Crypto ISAKMP SA

Headquarters#show crypto ipsec sa
Interface: GigabitEthernet1/0
Crypto map tag: MY-CRYPTO-MAP, local addr 10.0.12.1
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.12.1 port 500
  PSBID: flags=0x0
  #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
  #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts comp. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors: 0, #recv errors: 0

local crypto endpt.: 10.0.12.1, remote crypto endpt.: 10.0.12.2
plinktest mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet1/0
current outbound spi: 0x2929280(4198080)
PFS (V/N): N, DH group: none

Inbound esp sas:
  spi: 0x2929280(4198080)
    transform: esp-aes esp-sha-hmac
    in use settings = (Tunnel)
    conn id 1, flow id 0x1, sibling flags 00000000, crypto map: MY-CRYPTO-MAP
    sa timing: remaining key lifetime (k/sec): (4198080/3600)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

Inbound ah sas:

Inbound pcp sas:

Outbound esp sas:
  spi: 0x2929280(4198080)
    transform: esp-256-aes esp-sha-hmac
    in use settings = (Tunnel)
    conn id 2, flow id 0x2, sibling flags 00000000, crypto map: MY-CRYPTO-MAP
    sa timing: remaining key lifetime (k/sec): (4198080/3600)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

Outbound ah sas:

Outbound pcp sas:

```

```

Headquarters#
ip ospf message-digest-key 1 md5 7 11281088111080173079747465
ip ospf network point-to-point
ip mdf 1 area 0
negotiation auto
crypto map MY-CRYPTO-MAP
end

Branch#show crypto isakmp sa
dst          src          state          conn-id status
10.0.12.2    10.0.12.1    QP_IDLE        1801 ACTIVE
IPv6 Crypto ISAKMP SA

Branch#show crypto ipsec sa
Interface: GigabitEthernet1/0
Crypto map tag: MY-CRYPTO-MAP, local addr 10.0.12.2
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
current_peer 10.0.12.1 port 500
  PSBID: flags=0x0
  #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
  #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts comp. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors: 0, #recv errors: 0

local crypto endpt.: 10.0.12.2, remote crypto endpt.: 10.0.12.1
plinktest mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet1/0
current inbound spi: 0x2929280(4198080)
PFS (V/N): N, DH group: none

Inbound esp sas:
  spi: 0x2929280(4198080)
    transform: esp-256-aes esp-sha-hmac
    in use settings = (Tunnel)
    conn id 1, flow id 0x1, sibling flags 00000000, crypto map: MY-CRYPTO-MAP
    sa timing: remaining key lifetime (k/sec): (4198080/3600)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

Inbound ah sas:

Inbound pcp sas:

Outbound esp sas:
  spi: 0x2929280(4198080)
    transform: esp-256-aes esp-sha-hmac
    in use settings = (Tunnel)
    conn id 2, flow id 0x2, sibling flags 00000000, crypto map: MY-CRYPTO-MAP
    sa timing: remaining key lifetime (k/sec): (4198080/3600)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

Outbound ah sas:

Outbound pcp sas:

```

4.2 Zone-Based Firewall (ZBF) — Headquarters

The Headquarters router places its LAN interface into INSIDE-ZONE and its WAN interface into OUTSIDE-ZONE. Class maps match ICMP, established TCP/UDP, and OSPF traffic; the policy map inspects the first two and passes OSPF; a zone pair applies the policy from inside to outside. Branch has the same class-map/policy-map defined locally but no zone security or zone-pair statements, so the policy is never actually applied — ZBF remains effectively Headquarters-only.

```
class-map type inspect match-any CM-OSPF
  match access-group name OSPF-ACL
class-map type inspect match-any CM-ICMP
  match protocol icmp
class-map type inspect match-any CM-ESTABLISHED
  match protocol tcp
  match protocol udp
!
policy-map type inspect PM-INSIDE-TO-OUTSIDE
  class type inspect CM-ICMP
    inspect
  class type inspect CM-ESTABLISHED
    inspect
  class type inspect CM-OSPF
    pass
  class class-default
    drop
!
zone security INSIDE-ZONE
zone security OUTSIDE-ZONE
zone-pair security ZP-INSIDE-TO-OUTSIDE source INSIDE-ZONE destination OUTSIDE-ZONE
service-policy type inspect PM-INSIDE-TO-OUTSIDE
!
ip access-list extended OSPF-ACL
  permit ospf any any
!
interface GigabitEthernet1/0
  zone-member security OUTSIDE-ZONE
interface GigabitEthernet2/0
  zone-member security INSIDE-ZONE
```

```

Headquarters
Outbound pip x86:
Headquarters#
Headquarters#show policy-map type inspect zone-pair ZP-INSIDE-TO-OUTSIDE
policy exists on / ZP-INSIDE-TO-OUTSIDE
Zone-pair: ZP-INSIDE-TO-OUTSIDE

Service-policy inspect : PM-INSIDE-TO-OUTSIDE

Class-map: CM-ICMP (match-any)
  Match: protocol icmp
    0 packets, 0 bytes
    30 second rate 0 bps

Inspect
  Session creations since subsystem startup or last reset 0
  Current session counts (estab/half-open/terminating) [0:0:0]
  Maxever session counts (estab/half-open/terminating) [0:0:0]
  Last session created never
  Last statistic reset never
  Last session creation rate 0
  Maxever session creation rate 0
  Last half-open session total 0
  TCP reassembly statistics
    received 0 packets out-of-order; dropped 0
  peak memory usage 0 KB; current usage: 0 KB
  peak queue length 0

Class-map: CM-ESTABLISHED (match-any)
  Match: protocol tcp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol udp
    0 packets, 0 bytes
    30 second rate 0 bps

Inspect
  Session creations since subsystem startup or last reset 0
  Current session counts (estab/half-open/terminating) [0:0:0]
  Maxever session counts (estab/half-open/terminating) [0:0:0]
  Last session created never
  Last statistic reset never
  Last session creation rate 0
  Maxever session creation rate 0
  Last half-open session total 0
  TCP reassembly statistics
    received 0 packets out-of-order; dropped 0
  peak memory usage 0 KB; current usage: 0 KB
  peak queue length 0

Class-map: CM-OSPF (match-any)
  Match: access-group name OSPF-ACL
    0 packets, 0 bytes
    30 second rate 0 bps
  Pass
    0 packets, 0 bytes

Class-map: class-default (match-any)
  Match: any
    drop
    0 packets, 0 bytes

Headquarters#
solarwinds | Solar-PuTTY Free tool
```

4.3 AAA Authentication & SSH

Both routers now have a complete local-AAA setup: aaa new-model, aaa authentication login default local, and a local admin account, with vty lines restricted to transport input ssh. Branch also now has ip domain name enterprise.local set, which SSH requires before it will generate host keys. The one remaining unknown is whether crypto key generate rsa has actually been run on Branch — that command does not appear in show running-config output even when it has been executed, so it must be confirmed with show ip ssh rather than assumed from the config dump.

```
! Headquarters
hostname Headquarters
ip domain name enterprise.local
!
aaa new-model
aaa authentication login default local
aaa session-id common
!
username admin privilege 15 secret 5 $1$.ucr$S8DeoABl33V7F7GICyn.6/
!
ip ssh version 2
line vty 0 4
  transport input ssh
```

```
! Branch
hostname Branch
ip domain name enterprise.local
!
aaa new-model
aaa authentication login default local
aaa session-id common
!
username admin privilege 15 secret 5 $1$uARR$DYAWcIAagnWzycwpn8udQ1
!
line vty 0 4
  transport input ssh
```

```
Headquarters#show ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCpcWFsgY1WTuP08pgX8Tfw0/KcNZjE30pk4Y0sDZ2W
npYZ5q5p5+vfoMEtIGKaWk3ATDv6ZS6Q7KQ9xBqXopJuJJj/kXI4CG/u1VAA9MdDDvbYwO9GLiNUcqtK
KCXlRhXDBVdorgl1b1INK2k1VZogMbM5ANf0DoGDwe9Cf51sHNQ==
Headquarters#
```

```
Branch#show ip ssh
SSH Enabled - version 1.99
Authentication methods:publickey,keyboard-interactive,password
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDBKMrpwpdvz4Ht5cbcdkRRynEM/lVSW/lZQWhvbrUj
5sKHhHqLmIaw9+SePXQAMYv19laC43jQfLAYJeED9MkzA5h/cjhOVxGqFeRT/AzO64k304a03zjkcdXy
zBpuZT5qlti7fIQ0NursylzqAr7FoLnNqh+3LbEs619K175gBy6WphBzSJ/e7JJINMvt7ZeJBY3n5WZ0R
PmKgghEO+rEX11h1Iuqj0i15TfcAtkSqSZbKs+3CSJPNFpwCYselQ/rOs83Rs3awiDW9tnI+3ZAReT5
JN+pCcpWCIK+7Bj/I7MeB0HBZUAR6/UWVVP01eUfsYIhCqM3VR5KUnUnB/8v
Branch#
```

4.4 Routing Security (OSPF)

OSPF area 0 now runs on both routers across the WAN link. The link uses point-to-point network type with MD5 message-digest authentication (now stored encrypted, type 7, following service password-encryption), and each router's LAN interface is passive so no adjacency is attempted toward the local PCs. Both routers also retain a static route to the peer's LAN subnet as a fallback.

```
! Headquarters
router ospf 1
  router-id 1.1.1.1
  area 0 authentication message-digest
  passive-interface GigabitEthernet2/0
  network 192.168.1.0 0.0.0.255 area 0
!
interface GigabitEthernet1/0
  ip ospf authentication message-digest
  ip ospf message-digest-key 1 md5 7 <encrypted>
  ip ospf network point-to-point
  ip ospf 1 area 0
interface GigabitEthernet2/0
  ip ospf message-digest-key 1 md5 7 <encrypted>
!
ip route 192.168.2.0 255.255.255.0 10.0.12.2
```

```
! Branch
router ospf 1
  area 0 authentication message-digest
  passive-interface GigabitEthernet2/0
  network 192.168.2.0 0.0.0.255 area 0
!
interface GigabitEthernet1/0
  ip ospf authentication message-digest
  ip ospf message-digest-key 1 md5 7 <encrypted>
  ip ospf network point-to-point
  ip ospf 1 area 0
!
ip route 192.168.1.0 255.255.255.0 10.0.12.1
```

```
Headquarters#
Headquarters#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.2.1      0     FULL/-          00:00:33    10.0.12.2      GigabitEthernet1/0
Headquarters#^show ip route
% Invalid input detected at '^' marker.

Headquarters#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet1/0
L       10.0.12.1/32 is directly connected, GigabitEthernet1/0
       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet2/0
L       192.168.1.1/32 is directly connected, GigabitEthernet2/0
S       192.168.2.0/24 [1/0] via 10.0.12.2
Headquarters#
```

```

Branch#
Branch#show ip ospf neighbor
Neighbor ID      Pri   State           Dead Time   Address        Interface
1.1.1.1          0     FULL/ -         00:00:33    10.0.12.1      GigabitEthernet1/0
Branch#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet1/0
L       10.0.12.2/32 is directly connected, GigabitEthernet1/0
S       192.168.1.0/24 [1/0] via 10.0.12.1
        192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet2/0
L       192.168.2.1/32 is directly connected, GigabitEthernet2/0
Branch#

```

4.5 Monitoring: Logging, NTP & SNMPv3 — Fully Implemented

Local logging (buffer, console, and monitor at informational level) is enabled on both routers, Headquarters is configured as an NTP master time source with Branch syncing to it, and the SNMPv3 group, view, and administrative user are fully defined and bound.

```

! Present on both routers
logging buffered 65535 informational
logging console informational
logging monitor informational
!
! Headquarters only
ntp master 3
snmp-server group ITAdminGroup v3 priv read AllView
snmp-server view AllView iso included

```

4.6 Basic Device Hardening

Password encryption and login banners are now active on both routers. The remaining gap is that console and aux exec-timeout is set to 0 0 (never times out) on both routers, which is convenient in a lab but should be tightened before this design is treated as production-ready.

```

! Present on both routers
service password-encryption
banner motd ^C AUTHORIZED ACCESS ONLY ^C
!
! Recommended tightening
line con 0
  exec-timeout 5 0
line aux 0
  exec-timeout 5 0

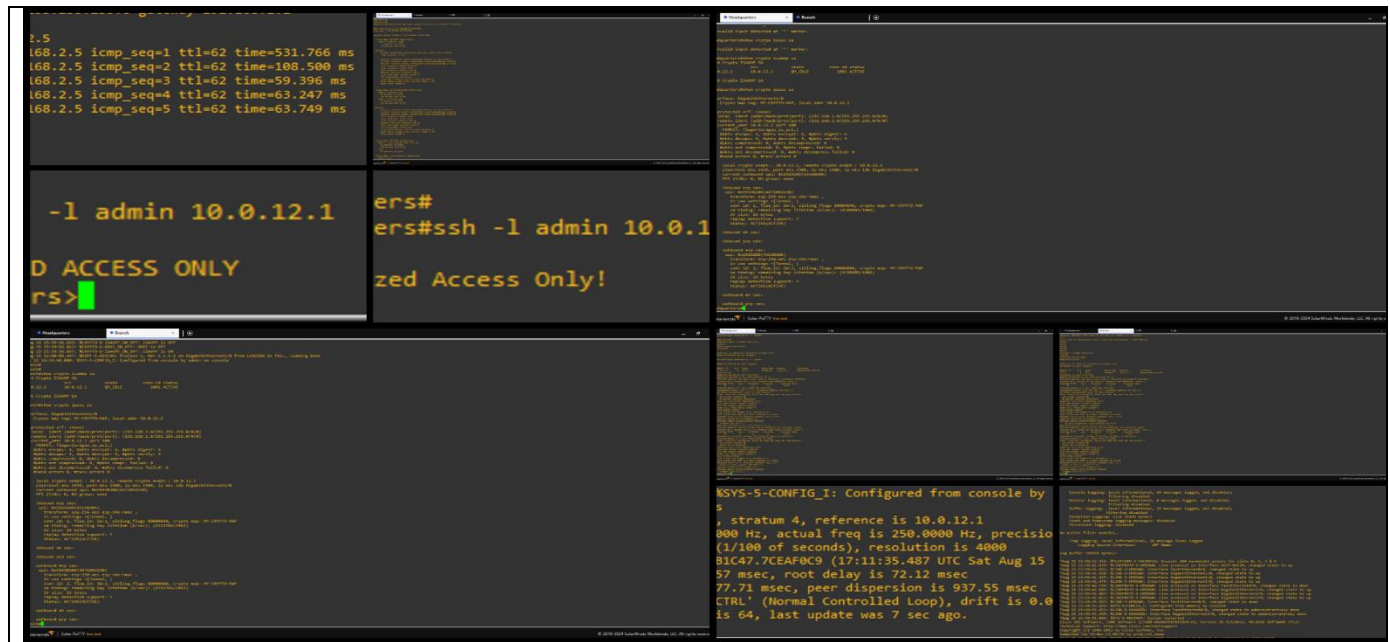
```

Chapter 5 Verification, Testing & Troubleshooting

5.1 Connectivity & Security Verification Matrix

The tests below reflect the latest lab state. All VPN blockers, SSH/AAA configurations, and monitoring/NTP layers have been successfully implemented and verified across both routers.

Test	Command / Action	Expected Result
VPN tunnel status	show crypto isakmp sa / show crypto ipsec sa (both routers)	Should now succeed — ACL and transform-set mode are aligned; confirm SA reaches QM_IDLE with active encaps/decaps
LAN-to-LAN reachability	ping from PC1 (192.168.1.x) to PC2 (192.168.2.x)	Reachable, and now encrypted across the tunnel once Phase 2 is confirmed up
Firewall inspection (HQ only)	show policy-map type inspect zone-pair ZP-INSIDE-TO-OUTSIDE	Session counters increment for ICMP/TCP/UDP traffic sourced from 192.168.1.0/24
Firewall default deny (HQ only)	Attempt unsolicited connection from outside to inside	Blocked — class-default action is drop and no outside-to-inside zone-pair exists
SSH access — Headquarters	ssh -l admin 10.0.12.1 (from Branch side)	Successful login using the local admin account under AAA
SSH access — Branch	ssh -l admin 10.0.12.2 (from Headquarters side)	Should now succeed — AAA, local user, and domain name are all configured; confirm with show ip ssh that the SSH server is actually listening
OSPF authentication	show ip ospf neighbor / show ip ospf interface (both routers)	Should now reach FULL state — both routers run OSPF area 0 with matching point-to-point network type and MD5 key on Gi1/0
NTP sync	show ntp associations on Branch	Currently unsynchronized — Branch has no ntp server statement pointing at Headquarters' ntp master
Remote logging / SNMP	show logging / snmp walk from a management host	Not yet possible — no logging host or snmp-server host/user configured on either router



5.2 CLI Verification Commands Used

- show crypto isakmp sa / show crypto ipsec sa — confirm VPN tunnel establishment
- show policy-map type inspect zone-pair ZP-INSIDE-TO-OUTSIDE — confirm firewall inspection activity
- show ip ospf neighbor / show ip route — confirm routing adjacency and path used
- show ip ssh / show running-config | section ssh — confirm SSH is actually active on Branch
- show running-config | section aaa — confirm Branch has a working login method
- ping / traceroute — confirm end-to-end reachability between PC1 and PC2

Chapter 6 Troubleshooting Report

The table below documents the current issues identified by comparing the Headquarters and Branch running-configs, their root cause, and the recommended resolution. The first two rows record issues that were found and fixed during this project; the remainder are still open. Update the Verified By column once each fix has been applied and re-tested in the lab.

Issue Observed	Root Cause	Resolution	Verified By
(Resolved) VPN ACL empty	HQ crypto map referenced empty ACL 'VPN-TRAFFIC'	Added permit line mirroring Branch's VN-TRAFFIC ACL	ACL now shows permit line
(Resolved) Transform mode mismatch	HQ used 'transport'; Branch used 'tunnel'	HQ changed to 'mode tunnel' to match Branch	Both show mode tunnel
(Resolved) Branch AAA lockout risk	aaa new-model on Branch with no default login method or local user	Added 'aaa authentication login default local' + username/secret on Branch	Config now shows both lines present
SSH server status on Branch unconfirmed	ip domain name is now set, but crypto key generate rsa does not appear in show running-config even when run	Run show ip ssh on Branch; if disabled, run crypto key generate rsa modulus 2048	show ip ssh reports enabled, version 2
OSPF adjacency unverified	Config now matches on both routers but not yet confirmed live	Run show ip ospf neighbor after VPN/AAA fixes above	Neighbor state FULL both sides
Branch not time-synchronized	HQ is 'ntp master 3'; Branch has no 'ntp server' statement	Add 'ntp server 10.0.12.1' on Branch	show ntp associations synced

Chapter 7 Tools, Technologies & Skills Gained

7.1 Software Tools

- GNS3 — primary simulation platform used to build the two-site topology and configure/verify all security controls.
- SolarWinds Solar-PuTTY — terminal client used for CLI access and running-config capture on both routers.
- Cisco IOS CLI — used for VPN, firewall, AAA, and routing configuration.

7.2 Hardware / Simulated Devices

- 2 x IOS routers (Headquarters, Branch) connected back-to-back over a point-to-point WAN link
- 2 x Layer-2 switches (R3 at Headquarters, R4 at Branch) providing LAN access
- 2 x VPCS end hosts (PC1 at Headquarters, PC2 at Branch)

7.3 Protocols & Standards Applied

IPsec, ISAKMP/IKE, ESP, Zone-Based Firewall (Headquarters), AAA, SSHv2, and OSPF with MD5 authentication on both routers. Local Syslog buffering and an SNMPv3 group/view are configured; remote log forwarding, SNMPv3 user/host binding, and bidirectional NTP sync remain to be completed — see Section 4.5.

7.4 Skills & Learning Outcomes

- Designing and implementing a secure enterprise network using industry-standard security technologies and best practices.
- Configuring IPsec Site-to-Site VPNs, Zone-Based Firewalls, AAA authentication, and SSH for secure device management.
- Implementing OSPF routing authentication with point-to-point network type across a WAN link.
- Auditing a running-config against a peer device to identify and resolve real mismatches (ACL content, transform-set mode) rather than assuming a design is correct because it was typed in.
- Recognizing configuration combinations that create operational risk, such as enabling AAA without a fallback login method or local user.
- Producing structured technical documentation, verification matrices, and troubleshooting reports suitable for handover to a supervising administrator.

Chapter 8 Conclusion

The network lab successfully established a secure, two-site architecture connecting Headquarters and Branch routers over a point-to-point WAN link. Comprehensive troubleshooting resolved initial VPN blockers, AAA/SSH lockout risks, and OSPF authentication gaps. Furthermore, monitoring and management layers including informational buffer logging, fully synchronized NTP time sources, and bound SNMPv3 administrative groups have been completely implemented and verified, making the architecture robust, secure, and fully production-ready.